

Lab: Mapping Your Creative System

Objective

Map the creative system in which you currently operate (or intend to operate) as an Active Inference system. By the end of this lab, you will have a concrete diagram of your creative system's Markov blanket, an inventory of its constraints and affordances, and an actionable plan for modifying the system to better support your inventive goals.

Materials / Prerequisites

- Your own invention idea or project (any domain, any stage)
- Paper and pen/pencil (or digital drawing tool)
- A timer
- Access to your actual workspace or a clear memory of it
- Completed reading of Module 01: The Creative System

Part 1: Define Your Creative System's Boundary (12 minutes)

Every creative system has a boundary — the Markov blanket — that separates its internal processes from the external environment. Your task is to identify this boundary for your own inventive practice. Think broadly about what constitutes your creative system. It is not just your desk or workshop — it includes your digital tools, your reference materials, the people you regularly consult, the websites you browse for inspiration, and the communities you participate in. The Markov blanket includes everything that mediates between your internal creative states and the external world.

Step 1 (4 min): Describe the physical and informational boundaries of your creative system. Where do you invent? What physical spaces, digital tools, and social networks constitute your system?

Write your response here

Step 2 (4 min): Identify all the **sensory inputs** — the channels through which information enters your creative system. Fill in the table below:

Input Channel	Type (Physical / Digital / Social)	Frequency (Daily / Weekly / Rare)	Quality (Rich / Adequate / Poor)

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Step 3 (4 min): Identify all the **active outputs** — the ways your creative system produces results. These include prototypes, sketches, writings, conversations, demonstrations, and any other externally-visible creative products.

Output Channel	Type	Frequency	Who Receives It?

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Part 2: Inventory Your Constraints (12 minutes)

Constraints are the priors of your creative system — they define the space of possible inventions. Most inventors think of constraints as obstacles, but the Active Inference framework reveals them as generative forces that focus the creative search space. A project with no constraints has no direction — every possibility is equally likely, which means no

possibility is salient. Constraints create the gradient that tells you which direction to explore. In this exercise, you will catalog all the constraints operating on your creative system and then practice the critical skill of reframing blocking constraints as focusing ones.

Step 1 (6 min): List every constraint on your creative system. Be thorough. Include constraints you normally take for granted.

Constraint	Category (Time / Money / Materials / Knowledge / Social / Regulatory)	Severity (Hard / Soft / Self-imposed)

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Step 2 (6 min): For each constraint, answer: Does this constraint **focus** your creative search (making it more productive) or **block** it (preventing valuable exploration)? Reframe at least two "blocking" constraints as "focusing" constraints by identifying what they force you to do differently.

Constraint 1 reframe:

Write your response here

Constraint 2 reframe:

Write your response here

What did you learn from the reframing exercise?

Write your response here

Part 3: Map Your Affordances (12 minutes)

Affordances are the action possibilities your creative system offers. James Gibson's concept of affordances — what the environment offers to an agent — maps directly to the Active Inference concept of available policies. Your creative system's affordances determine the adjacent possible: the set of inventions you could create next given your current tools, materials, knowledge, and relationships. An inventor with access to a 3D printer, a soldering station, and a machine shop has a different adjacent possible than one with only a laptop and

internet access. Neither is better — they simply afford different types of invention. This exercise maps your current affordances and identifies gaps you might want to fill.

Step 1 (4 min): Walk through your workspace (physically or mentally). List every tool, material, resource, and relationship that enables creative action:

Write your response here

Step 2 (4 min): Now consider your "adjacent possible" — what inventions or invention steps are within reach given your current affordances? List at least five things you could prototype or test with your existing resources, without acquiring anything new:

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Step 3 (4 min): Identify three affordances that are **missing** from your creative system — action possibilities that would significantly expand your adjacent possible. For each, note how you might acquire it:

Missing Affordance	What It Would Enable	How to Acquire It

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Part 4: Analyze Your Feedback Loops (12 minutes)

Feedback loops are the engine of creative progress. Every time you build something, observe the result, update your understanding, and build again, you are completing one cycle of the perception-action loop that drives Active Inference. The speed and richness of these loops determine how quickly your creative system can learn and improve. Some loops are fast (you build a sketch and immediately see whether the proportions work); others are slow (you launch a product and wait months for market feedback). Understanding which loops are operating and how quickly they cycle gives you leverage to accelerate your creative process.

Step 1 (6 min): Identify the feedback loops currently operating in your creative system. For each loop, note how quickly it cycles and what information it carries:

Feedback Loop	Cycle Time	Information Carried	Effect on Creativity
Example: Build something, see if it works	Hours to days	Does the prototype function?	Rapid iteration on design

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Step 2 (6 min): Draw a diagram of your creative system showing: - The boundary (Markov blanket) as an outer shape - Internal components (you, tools, materials, knowledge bases) - Sensory inputs as arrows entering from outside - Active outputs as arrows leaving - Feedback loops as curved arrows connecting outputs back to inputs - Constraints as labels on the boundary

Upload or describe your diagram here:

Write your response here

Part 5: Design Intervention (12 minutes)

The purpose of mapping a creative system is not academic — it is to identify specific, actionable modifications that will improve the system's inventive capacity. In Active Inference terms, you are acting on your own creative system to reduce the free energy between what it currently produces and what you want it to produce. The most effective interventions target the aspect of the system that is currently the biggest bottleneck: a missing sensory input (you

are not getting important information), a blocking constraint (a rule that prevents productive exploration), a missing affordance (a tool or relationship you need), or a broken feedback loop (you are building without testing, or testing without learning). Based on your analysis in Parts 1 through 4, design one concrete modification.

The modification I will make:

Write your response here

Which aspect of the system it changes (boundary / constraints / affordances / feedback loops):

Write your response here

Why I expect this to improve creative output (in Active Inference terms):

Write your response here

How I will measure whether the modification worked:

Write your response here

Timeline for implementing this modification:

Write your response here

Discussion / Debrief Questions

1. Which element of your creative system analysis surprised you most? Was it an input you did not realize you were relying on, a constraint you had not recognized, or an affordance you were not using?
2. Compare your creative system to one of the case studies from the module reading (Bell Labs or Shenzhen). What structural feature do they have that your system lacks? Is it feasible to add it?
3. How does the distinction between "smooth" and "rugged" free energy landscapes apply to your current creative system? Is your system better suited for incremental or radical innovation? Is that what you need?
4. If you were designing a creative system from scratch for your invention project — with unlimited resources — what would it look like? How does it differ from your current system, and what is the most important single difference?
5. Consider the relationship between your creative system and the broader ecosystem (your industry, your community, the state of technology). How does the broader system constrain or enable what your creative system can achieve?